

LABORATORY ACTIVITY I

Introduction to Biomechanics

Observing the Basis of Motion: An Introductory Biomechanical Analysis of Human Movement

I. Learning Outcomes

- ✓ Analyze the basis for human movement based on the definitions of mechanics, biomechanics, and kinesiology.
- ✓ Understand and explain the importance of biomechanics to movement, exercise, and sport.
- ✓ Apply biomechanical knowledge and information to the observation and interpretation of human movement.

II. Pre-Laboratory Questions

1. In your own words, define mechanics, biomechanics, and kinesiology, and explain how the three relate to one another.
2. Name the three anatomical planes of motion and give one example of a human movement that occurs primarily in each.
3. Why might a coach or educator rely on visual, qualitative observation instead of, or alongside, expensive motion-analysis equipment?
4. What is qualitative movement analysis, and why does trained human observation still matter even as AI-based motion analysis tools become more common?

III. Introduction

Human movement can be studied from several angles, and the terms used to describe those angles are often used loosely in everyday conversation. Mechanics is the branch of physics concerned with forces and the motion they produce, traditionally divided into statics (forces on bodies at rest or in equilibrium) and dynamics (forces on bodies in motion, further split into kinematics, which describes motion without regard to its cause, and kinetics, which explains the forces that cause it). Biomechanics applies these same mechanical principles to living organisms, and in the context of human movement, it sits directly at the meeting point of engineering mechanics and biological science.

Kinesiology is the broader academic discipline, encompassing not just the mechanics of movement but also anatomy, physiology, motor learning, and the psychosocial dimensions of physical activity. Biomechanics is therefore best understood as one of kinesiology's foundational sub-disciplines: the lens through which movement is measured, described, and explained in terms of forces, levers, and motion. A useful working distinction for this laboratory is that mechanics supplies the laws, biomechanics applies those laws to the moving human body, and kinesiology places that application within the wider study of why and how people move.

Scientific Principles

This laboratory is anchored on three interlocking frameworks:

- Newton's Three Laws of Motion — inertia, acceleration ($F = ma$), and action-reaction — which explain why the body resists changes in motion, why greater force applied to a segment produces greater acceleration, and why every push against the ground is met with an equal ground reaction force.

- Kinematic versus kinetic description — kinematics covers displacement, velocity, joint angle, and cadence; kinetics covers the forces, torques, and muscular effort producing that motion. Observation-based labs work primarily at the kinematic level, since force cannot be seen directly, only inferred from its visible effects.
- The anatomical planes and axes system (sagittal, frontal, transverse) together with the concepts of center of gravity (COG) and base of support (BOS), which give observers a shared vocabulary and reference frame for describing where and how a movement is occurring.

Practical Significance

These principles are not abstract. Coaches, physical educators, and rehabilitation professionals rely on biomechanical reasoning daily to refine technique, prevent injury, and design conditioning programs. Editorial work in the field has pointed to the expanding role biomechanics now plays in three connected areas: optimizing sport performance, reducing injury risk, and guiding athlete rehabilitation, particularly as computer-assisted motion analysis and wearable sensors become more accessible to practitioners working outside elite sport-science laboratories.

At the same time, the skill this lab targets structured visual observation remains the entry point into all that technology. Even scoping reviews tracking the rise of AI-assisted motion analysis and injury prediction note that these computational tools are trained against, and ultimately still checked by, the trained eye of a coach or clinician reading movement quality in real time. Before a student ever touches a force plate or a markerless motion-capture app, the capacity to watch a squat, a gait cycle, or a throw and correctly name what is happening biomechanically is the more fundamental and more transferable skill.

Current Evidence

Recent biomechanics-education literature reinforces the design of this laboratory in three ways:

- Tiwari, Sinha, Prasad, Badhyal, and Joseph (2024) reviewed active-learning strategies for introductory biomechanics courses taught in low-resource settings, and found that experiential, hands-on laboratory tasks — rather than lecture-only delivery — measurably strengthen students' grasp of biomechanical concepts even without commercial motion-analysis software or equipment. This directly supports an observation-first, low-instrumentation lab design of the kind used here.
- Dhahbi (2025), introducing a special research collection on biomechanics, described the field's growing contribution to performance optimization, injury-risk mitigation, and rehabilitation across athlete populations, while flagging that longitudinal and applied classroom-level work is still needed to translate laboratory findings into everyday coaching and teaching practice.
- Souaifi et al. (2025), in a scoping review of artificial intelligence applications in sports biomechanics spanning studies from 2015 to 2024, reported that computer-vision and machine-learning tools are increasingly used for motion analysis and injury prediction, but also noted that these systems are calibrated against, and evaluated relative to, expert human judgment of movement — underscoring why foundational qualitative observation skills remain essential training for future practitioners even in a technology-rich field.

a. Learning Outcomes

- Operationally define and correctly apply core biomechanical terms (kinematics, kinetics, plane of motion, axis, center of gravity, base of support) to a live, observed movement.

- Differentiate mechanics, biomechanics, and kinesiology using the same observed movement as a shared reference point.
- Describe, in practical terms, how biomechanical observation informs coaching, teaching, or clinical decision-making.

b. Scenario

You have been assigned as a student intern supporting a PATHFIT instructor / strength-and-conditioning coach during a pre-activity movement screening. Before your athletes begin a new training block, you are asked to observe two everyday movement tasks such as walking gait and an overhead squat, and describe what you see using biomechanical language. No motion-capture cameras or force plates are available; your only tools are a stopwatch, a tape measure, a goniometer, a plumb line, and a trained eye. Your observations will be used to flag any athlete who may benefit from technique coaching or a referral for further assessment.

c. Key Biomechanical Concepts (Operational Definitions)

Term	Operational Definition for the Activity
Mechanics	The branch of physics dealing with forces and the motion they produce in any object, living or non-living.
Biomechanics	The application of mechanical principles to describe and explain the forces acting on, and produced by, the human body during movement.
Kinesiology	The academic discipline studying human movement in its full scope — mechanical, anatomical, physiological, and behavioral.
Kinematics	The description of motion in terms of position, displacement, velocity, and joint angle, without reference to the forces causing it.
Kinetics	The study of the forces (muscular, gravitational, ground-reaction) that cause or modify motion.
Plane of motion	One of three imaginary flat surfaces (sagittal, frontal, transverse) that divide the body and along which movement is described.
Axis of rotation	The imaginary line around which a body segment rotates during a joint movement.
Center of gravity (COG)	The point at which the body's mass is considered to be balanced in all directions.
Base of support (BOS)	The area beneath a person bounded by all points of contact with the supporting surface.
Qualitative movement analysis	The systematic, non-numerical observation and judgment of movement quality, typically using a structured checklist.

IV. Laboratory Preparations

Venue: High Performance Gym

- A clear, level walkway of at least 6 meters for the gait-observation task.
- Adequate, even lighting so that joint positions and body alignment are clearly visible to observers.

- Enough open floor space for one demonstrator to perform an overhead squat while at least two observers view from the side (sagittal) and front (frontal) simultaneously.
- A quiet enough environment that verbal cues and stopwatch timing can be heard clearly.

V. Equipment and Materials

- Masking tape or floor cones (to mark the walkway and a fixed observation point)
- Measuring tape
- Stopwatch or smartphone timer
- Smartphone or tablet camera, for optional slow-motion playback review (with participant consent)
- Goniometer (for baseline joint-angle checks)
- Plumb line or a string with a small weight attached (to visualize vertical alignment of the center of gravity)
- Rating Sheets, clipboards, and pens
- Blank paper for data entry
- Comfortable athletic wear for the student who volunteers as movement demonstrator

VI. Safety Considerations

- All demonstrators must complete a brief dynamic warm-up before performing any observed movement.
- Inspect the walkway and squat area for obstructions, loose tiles, or slippery spots before starting.
- Demonstrators must stay within their own comfortable, pain-free range of motion — no forced stretching or maximal-effort attempts for the sake of a better observation.
- Rotate the demonstrator role among group members to prevent fatigue-related movement changes.
- If any video recording is used, obtain the demonstrator's verbal consent first, and delete or exclude footage of anyone who declines.
- Stop the activity immediately if a demonstrator reports pain, dizziness, or discomfort, and inform the instructor.

VII. Data Collection Procedure

Because this laboratory uses observation rather than instrumented measurement, precision and accuracy depend on discipline in method rather than on equipment:

- Assign fixed observer positions (one directly to the side for sagittal-plane viewing, one directly in front for frontal-plane viewing) and keep those positions constant across trials.
- Collect at least three trials per movement task (three walking passes, three squat repetitions) and record each trial separately before averaging, rather than relying on a single impression.
- Use the goniometer and plumb line to check joint angles and vertical alignment to the nearest 5 degrees; do not estimate angles by eye alone when the tool is available.
- Count cadence over a fixed interval (15 seconds) and multiply by four, rather than counting an unclear or variable time span.
- If filming for a second look, keep the camera distance, height, and angle identical across trials so that frames are directly comparable.
- Where two or more observers rate the same trial, compare ratings and discuss any disagreement before recording a final value, to reduce single-observer bias.

References

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Laboratory Sheet Activity I

Date : _____
 Week Number : _____
 Laboratory Number : _____
 Student Name(s) : _____

 Section: : _____
 Duration: _____

I. Data Collection

a. Data Collection Tables

Table 1. Walking Gait Observation

Trial	Cadence (steps/min)	Stride Length – L (m)	Stride Length – R (m)	Arm Swing Symmetric? (Y/N)	Trunk Lean Present? (Y/N)
1					
2					
3					
Average					

Table 2. Overhead Squat Observation Checklist

Criterion	Present / Absent	Primary Plane Involved	Observations
Heels remain in contact with the floor			
Knees track in line with the toes (no excessive valgus/varus)			
Trunk remains upright with a neutral spine			
Arms remain overhead throughout the descent and ascent			
Left-right symmetry maintained (hips, knees, arms)			

b. Calculations

- Average Cadence = (steps counted in 15 seconds) × 4 = _____ steps/min
- Average Stride Length (Right side) = sum of trial values/number of trials = _____ m
- Average Stride Length (Left side) = sum of trial values/number of trials = _____ m
- Symmetry Index (%) = $\frac{|\text{Left value} - \text{Right value}|}{[(\text{Left value} + \text{Right value}) \div 2]} \times 100 =$ _____ %
- Show your computation work in the space below:

c. *Graphs and Motion Analysis*

In the space below, sketch a simple stick-figure diagram of the overhead squat at its lowest point (side view). Label the hip, knee, and ankle joint angles you measured with the goniometer, and mark the estimated line of the center of gravity using your plumb-line reading.

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d. Interpretation Questions

1. Based on your gait data, was the walking pattern kinematically efficient? Justify your answer using the cadence, stride length, and symmetry values you recorded.

2. Which biomechanical principle best explains any trunk lean or left-right asymmetry you observed?

3. Identify the primary and secondary planes of motion involved in the overhead squat, citing specific joint actions you observed.

4. How would these observations inform a coach's or instructor's decision about whether a corrective intervention is needed?

II. Application

Using your recorded data, propose one coaching cue or corrective exercise that addresses a movement inefficiency you observed in either task. Justify your recommendation by naming the specific biomechanical principle it addresses.

II. Reflection

1. What was the most challenging part of applying biomechanical observation without instrumented equipment?
2. How has this activity changed, or sharpened, your understanding of the relationship between mechanics, biomechanics, and kinesiology?
3. In what future professional situation — teaching, coaching, or rehabilitation — do you see

III. Assessment Rubric

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Accuracy of data collection	All trials recorded precisely and consistently using tools and procedure specified	Minor inconsistency in one trial or measurement	Data collected but procedure not consistently followed	Data missing, incomplete, or collected without following procedure
Use of biomechanical terminology	Terms used correctly and precisely throughout	Terms mostly correct, minor misuse	Terms used but with noticeable errors	Terms largely absent or incorrect
Depth of interpretation	Interpretation clearly links data to biomechanical principles	Interpretation mostly linked to principles, some gaps	Interpretation present but superficial	Interpretation missing or unsupported by data
Quality of application/recommendation	Recommendation specific, justified, and biomechanically sound	Recommendation reasonable, justification adequate	Recommendation vague or weakly justified	Recommendation absent or not justified
Reflection quality	Thoughtful, specific, and personally grounded reflection	Adequate reflection with some specificity	Reflection generic or brief	Reflection missing or off-topic

Total Score: _____ / 20